



DEZAI

Web Platform Engineering Manual & AWS Cloud Architecture Specification

An extended production blueprint mapping out decoupled web delivery interfaces, real-time proctor networks, localized generative learning models, and an enterprise-grade AWS Cloud architecture supporting massive high-availability scaling.

Document Architecture:	Volume XII (Cloud Infrastructure & Master Spec)
Frontend Infrastructure:	Next.js SSR Web Panels & Vercel Edge Networks
Backend Environment:	AWS EKS Cluster Containerization & Node.js Core
Cloud Target Stack:	AWS Lambda, Amazon Aurora PostgreSQL, AWS Bedrock
Target Production Date:	August 10, 2026

1. EXECUTIVE VISION & THE CROSS-INSTITUTIONAL LEDGER

Dezai AI is establishing a next-generation, web-native microcredential ecosystem engineered to dissolve conventional institutional friction and establish a unified, verified cross-university academic framework. By replacing legacy paper pathways with decentralized digital credentials, immutable competency badges, and embedded generative AI learning environments, the platform defines a transparent benchmark of verifiable student capability.

This document serves as the absolute blueprint for our engineering, design, and operations teams. It focuses heavily on functional splits, exact client-server interface duties, state engines, and visual rules, intentionally excluding raw syntax blocks to remain a conceptual engineering mandate. The ecosystem is optimized for global access, shifting heavy multimedia workloads onto client devices through intelligent typographic layouts.

1.1 Core Institutional Interconnection Model

To establish structural trust across independent academic networks, the Node.js API perimeter isolates three core communication flows. This network bypasses manual registry processes via programmatic validation rules:

- **Universal Microcredential Ledger:** Every completed course or skill badge is broken down into structured, atomic criteria. The registry records verification keys, syllabi definitions, passing parameters, and anonymous user hashes.
- **The Unified Network Core:** Decoupled student routing registries ensure that a user from any partner university can access resources natively. The database platform matches identical syllabi configurations across physical boundaries, ensuring that student competencies are judged using standard grading criteria.
- **Secure API Communication Perimeter:** Edge network routes leverage token-based handshakes to query, parse, and coordinate course completions within external university registrar databases without exposing raw private data rows or writing complex integrations for outdated institutional ERP tools.

2. COURSE ACCESSIBILITY & THE TRI-ACCESS WEB FRAMEWORK

To drive global adoption, prevent institutional monopolies, and unlock continuous pathways for students across varying socioeconomic demographics, the backend web routing layer enforces three deterministic gateway access controllers.

2.1 Premium Flat-Rate Payment Pathway

The classical monetization loop processes direct checkouts. When an authorized user executes a course purchase on the main Next.js interface, the web layer tokenizes payment attributes and handles transit securely via REST API to the Node.js environment. The backend interfaces with external card network gateways using secure tokenized hooks. Upon receipt of verified webhooks, the core engine logs an absolute access permission parameter onto the student's cloud metadata profile, instantly releasing the learning blocks.

2.2 Cumulative Experience Point (XP) Accumulation Ledger

To incentivize continuous learning and provide a zero-cost option for highly motivated individuals, the platform coordinates a value-exchange economy driven entirely by student interaction:

- **Granular Earning Metrics:** The server-side transaction logging framework increments user point values automatically when a student completes micro-learning blocks, completes structural summaries, provides helpful community insights, or maintains active daily learning streaks.
- **Automatic Threshold Evaluation:** Premium tracks have a predefined, static entry point value. When the student selects the "Unlock via XP Ledger" trigger, the backend issues an isolated database transaction. If the balance satisfies the threshold, the server clears the gate and logs the access authorization token.

2.3 Institutional Leaderboards & Monthly Merit Grants

To stimulate healthy peer competition, the web interface coordinates real-time sorting tables that filter student performance by their verified university campus. At exactly 23:59:59 UTC on the final day of each calendar month, automated database triggers freeze the leaderboard states. The system isolates ranks 1 through 5, automatically generating zero-cost premium scholarship vouchers to guarantee access to top-tier microcredentials for elite performers without requiring financial checkouts.

3. HYBRID CONTENT DELIVERY ENGINE ARCHITECTURE

To scale rapidly across infrastructure-limited geographies while maintaining academic credibility, Dezai bypasses the massive hosting and bandwidth liabilities of video-only platforms through a highly optimized, hybrid content delivery architecture.

3.1 Core High-Impact Videos

Video streaming pipelines represent substantial operational costs and present high latency friction across remote regional cell sites. To mitigate this structural bottleneck, video production is reserved strictly for high-level conceptual pillars, architectural overviews, and introductory course frameworks. Content is recorded by respected local faculty members and high-achieving alumni selected via departmental recommendations and student polls. This maintains deep continuity with the student's active university curriculum.

MEDIA STORAGE & DISTRIBUTION FRAMEWORK

All high-impact media segments are processed into highly compressed chunks and deployed onto edge cache infrastructure. The client-side streaming wrapper optimizes bitrates dynamically based on connection quality, ensuring uninterrupted delivery without buffering delays.

3.2 Architectural Benefits of Hybrid Media Delivery

By splitting courses into sparse core videos and rich programmatic text components, the platform avoids the bandwidth bottlenecks common to legacy corporate training tools. This design choice prevents data delivery drops, reduces media pipeline dependencies, and lets engineering teams focus on improving the performance of responsive client-side interface elements.

4. THE EXPRESSIVE TYPOGRAPHIC ENGINE

For deep technical instruction, step-by-step code breakdowns, mathematical proofs, and secondary concepts, the platform shifts to a lightweight, typographic text model inspired by the variable layout principles of Geronimo Stilton publications.

4.1 The Visual Anchor Effect & Dynamic Mnemonics

Words are rendered client-side using dynamic typography that mirrors their exact functional behavior. This creates an immediate cognitive connection for complex concepts:

- **Memory Leak Simulation:** Text elements defining system resource depletion and uncollected heap addresses will physically drip, fade, and dissolve on the user's screen. This models real-world memory exhaustion behaviors directly within the reading material.
- **Overfitting Squeeze Component:** Mathematical explanations of overfitting in deep neural networks are automatically compressed into a rigid, non-adaptive layout border. This visual layout reinforces how models fail to generalize to broader data dimensions.
- **Interactive Cognitive Anchors:** Key architectural definitions alter font weights, morph color properties, or pulse upon viewport scrolling and cursor interaction, creating lasting visual associations.

4.2 The 99% Infrastructure Cost Reduction Framework

Instead of distributing gigabytes of raw, compressed 1080p video files over expensive CDN pipelines, the platform transmits lightweight, highly structured JSON/Markdown strings. The local frontend device processes and animates the expressive typography natively. This slashes hosting and data transfer costs by approximately 99%, guaranteeing instantaneous interaction even on legacy cellular networks in remote Tier-3 regions.

5. HIGH-INTEGRITY AUTOMATED EVALUATION ARCHITECTURE

To completely eliminate academic dishonesty and ensure corporate recruiters treat Dezai credentials with absolute trust, all courses terminate in a highly secure, time-proctored, fully automated quiz layer.

5.1 The 100:15 Dynamic Question Bank Engine

The core evaluation layer operates on an asymmetric randomization pattern designed to eliminate user collusion and answer-sharing networks completely:

- **The Database Matrix:** Partner faculty members and the internal editorial team seed each microcredential track with a robust pool of a minimum of 100 heavily vetted, multi-variant questions containing dynamic variables.
- **Algorithmic Shuffling & Sampling:** When a quiz instance initializes, the server-side logic layer samples exactly 10 to 15 randomized questions from the active database array.
- **Collusion Elimination:** Questions, option vectors, and internal formula variables are shuffled programmatically per unique session. Two users taking the exam side-by-side receive distinct question combinations and shuffled options, blocking answer leakage, peer-to-peer collusion, and systemic cheating.

5.2 Hardened Frontend Security (Anti-Copy/Paste Layer)

The client interface uses strict event controllers to isolate the testing canvas from outside applications. The browser canvas catches and halts all execution variants of clipboard manipulation tools. Students are locked from executing copy commands on test text blocks, preventing them from feeding questions into external search engines or generative AI tool interfaces. Input fields reject external pasted content, forcing users to write all answers manually.

6. REAL-TIME PROCTORING & EXAM STATE BOUNDARIES

Beyond question randomization and clipboard blocks, the quiz terminal enforces continuous behavioral analysis to maintain strict examination integrity across unproctored locations.

6.1 Focus-Loss Logging & Tab Switch Prevention

The examination application monitors page visibility states via native web browser hooks. If a student moves focus away from the active evaluation window, opens a parallel tab, or minimizes the testing frame, the proctoring engine logs the event instantly:

Violation Threshold	System Automation Response Action	User Interface State
First Offense	Halts the quiz countdown timer and triggers a prominent warning modal on the viewport. Logs the initial infraction timestamp to the backend.	Yellow Alert Border Overlay
Second Offense	Records a secondary flag, deducts 15% from the total available points, and locks the screen for 30 seconds to enforce focus.	Orange Flash Alert Banner
Third Offense	Terminates the test session immediately, discards all progress rows, and records a zero grade score onto the persistence database.	Red Exclusion Frame Screen

6.2 Time-Bounded Sprints & Strict Retake Constraints

Quizzes apply tight countdowns per item, removing the luxury of time required to lookup complex computer science principles externally. Students receive exactly three free attempts per module quiz. If they fail to clear the passing criteria within these cycles, the backend locks the route. To open a subsequent attempt, the student must clear an anti-spam interaction fee, protecting system resources and encouraging review before retaking tests.

7. DEEP GENERATIVE AI MODEL INTEGRATION LAYER

A core feature of the Dezai AI platform is the direct integration of a state-of-the-art Generative AI Model layer. This agent acts as an automated, multi-tiered personalized mentor, embedded natively within the Next.js interface to assist students in navigating, breaking down, and completing courses.

7.1 Context-Aware Course Navigation & Learning Paths

The AI model acts as an intelligent router for the student's learning path. Instead of forcing users through static folders, students can query the interface in plain language regarding their specific academic goals or technical concept gaps. The AI model reads the student's active progress metrics, past quiz performance, and course syllabi metadata via secure backend endpoints to map a custom sequence of learning nodes, pointing out exactly which sections to study next.

7.2 Automated Remediation During Exam Lockouts

When a student encounters a 24-hour quiz lockout following an examination failure, the AI model automatically switches to an intense remediation mode. The background Node.js engine extracts the anonymized error profile of the student's failed test answers and constructs a targeted context window for the AI model. The web interface then presents an active, conversational study sandbox where the AI explains the underlying mathematical or logical theories, presents interactive challenges, and ensures the student completely masters the material before the exam terminal unlocks.

7.3 Adaptive Summary Generation & Low-Bandwidth Notes

To support diverse learning speeds, the AI system handles automatic content translation. It reads dense microcredential video transcripts and heavy textbook files to extract clear, low-bandwidth text summaries, visual workflow outlines, and contextual concept flashcards. Students can adjust the complexity of these explanations on their browser dashboard, changing settings from high-level summaries to deep, granular technical analyses on the fly.

8. FULL FRONTEND VS. BACKEND FUNCTIONAL SPLIT MATRIX

To maintain crisp separation of concerns and enable concurrent development paths, the web architecture cleanly splits operations between the client-side UI and server-side logic engines.

8.1 Frontend Client Architecture Responsibilities

The Next.js framework renders client components to minimize processing overhead on our central servers. The browser application handles layout painting and interactive workflows:

- **State Management:** Maintains client tracking via lightweight store contexts to manage theme properties, live ranking shifts, examination tickers, and chat frames.
- **UI Responsiveness:** Employs structured utility styles to achieve fluid page adjustments across phone screens, tablet panels, and desktop interfaces.
- **Data Ingestion:** Dispatches secure asynchronous network calls to the backend infrastructure, handling data transmission through structured JSON formats and executing client-side route guards based on returned auth keys.

8.2 Backend Server Architecture Responsibilities

The server infrastructure operates as the authoritative brain of Dezai AI. It handles calculations, identity validation, and direct data safety boundaries:

- **Identity Control & Perimeter Security:** Manages authentication token validation, evaluates CORS policy configurations, and secures administrative data tables from external exposure.
- **Business Processing services:** Computes grading percentages, handles leaderboard calculations, processes point balance ledger rows, and enforces the 24-hour exam lockouts.
- **Persistent Storage Control:** Runs transactional object-relational mappings to coordinate state persistence across centralized cloud databases, utilizing strict data schemas to guarantee transactional integrity.

9. AWS CLOUD INTEGRATION & CORE INFRASTRUCTURE

To transition the Dezai AI platform from local development into an elastic, enterprise-grade application capable of supporting hundreds of thousands of concurrent users across universities, the entire system is hosted on an integrated Amazon Web Services (AWS) cloud architecture.

9.1 The AWS Core Architecture Layout

- **Compute & Container Routing Layer (AWS EKS & EC2):** The backend Node.js core microservices are packaged inside Docker containers and managed by an Amazon Elastic Kubernetes Service (EKS) cluster. EKS coordinates compute nodes dynamically across multiple Availability Zones (AZs) for instant failover coverage. Highly dynamic, event-based processes are run via AWS Lambda functions to reduce baseline server expenses.
- **Global Content Acceleration & Web Delivery (AWS CloudFront & S3):** Static assets, compiled font schemas, and core conceptual videos are safely stored within an encrypted Amazon S3 bucket. An Amazon CloudFront Distribution acts as our global Content Delivery Network (CDN), caching content at edge locations to ensure sub-50ms paint times for global users.
- **Authoritative Data Persistence (Amazon Aurora Serverless PostgreSQL):** Student profiles, course configurations, examination matrix records, and point balance ledgers are maintained inside a highly responsive Amazon Aurora relational database. The engine auto-scales computing capacity in response to testing peak loads and runs background read-replicas to protect main tables during heavy dashboard analytical queries.

AI ENGINE INTEGRATION (AWS BEDROCK & OPENSEARCH)

Our contextual generative mentoring system communicates directly with Amazon Bedrock via secure IAM access keys. Course metadata transcripts are tokenized and organized inside an Amazon OpenSearch Serverless database vector cluster to handle ultra-low latency context assembly.

10. AWS CLOUD MIGRATION STEPS & STRATEGIC EXECUTION

Transitioning the infrastructure to AWS requires a highly programmatic approach to ensure zero service disruption during active university semesters. The migration is broken down into structured, actionable phases.

10.1 Step-by-Step Infrastructure Deployment Path

- 1. Network & Perimeter Layer Initialization (VPC & IAM Roles):** Define an isolated Amazon Virtual Private Cloud (VPC) with segregated public and private subnet pairs across distinct data centers. Configure strict AWS Identity and Access Management (IAM) permission policies to follow the principle of least privilege.
- 2. Data Layer Setup & Synchronization (Aurora & S3 Migration):** Provision the Amazon Aurora Serverless database instance within the private subnet. Run secure data validation mappings to seed university information, question matrices, and course tables into the relational layer. Set up Amazon S3 buckets with custom access blocks.
- 3. Containerization & Orchestration Deployment (EKS & Lambda Setup):** Pack the Node.js API services into clean runtime containers and push them up into the Amazon Elastic Container Registry (ECR). Deploy these containers onto the EKS cluster, linking an AWS Application Load Balancer (ALB) to handle inbound user traffic safely.
- 4. Edge Routing & Secure Caching Configuration (CloudFront & Route 53):** Point our top-level domain records through Amazon Route 53. Spin up an AWS Certificate Manager (ACM) SSL script to provision end-to-end transport encryption. Connect CloudFront back to the static S3 storage buckets and the ALB backend endpoints.

10.2 Continuous Cloud Governance & Performance Tracking

To control infrastructure expenditures during weekends and holidays, the DevOps team sets up automated AWS Budgets alert flags. Amazon CloudWatch streams continuous performance diagnostics, analyzing CPU workloads, database query duration spikes, and error code tracking to trigger automated self-healing compute steps before users experience dashboard slowdowns.

11. ORGANIZATIONAL STRUCTURE & TEAM OPERATIONAL MATRIX

The execution of the Dezai AI web platform roadmap requires precise boundaries of responsibility across the entire staff roster. Below is the operational matrix mapping tasks to specific internal teams.

Team Role Scope	Daily Web System Operations & Scope	Critical Deliverable Targets
Frontend UI Engineers	Construct responsive web layouts within Next.js. Implement reactive dashboards, time-proctored quiz presentation nodes, and conversational chat shells. Integrate JSON/Markdown animation handlers for the Expressive Typographic Engine.	• Next.js layout trees • Local state store files • Typographic design styles
Backend Core Engineers	Build the Node.js application service layers. Configure API endpoints, manage relational database connections, and handle external secure payment key logic. Develop the dynamic quiz shuffling algorithm.	• Node.js routing trees • Relational database schemas • Question randomization layers
AI Integration Specialists	Manage external AI models and coordinate context prompting windows. Connect API keys via AWS Bedrock, tune vector embeddings within OpenSearch, optimize token usage costs, and build custom fallback mechanisms.	• Context window strategies • Ingestion pipelines for summaries • Bedrock connection layers
DevOps & Cloud Engineers	Provision isolated AWS VPC networks, configure EKS cluster nodes, manage Aurora scaling rules, set up IAM permission guards, maintain global CloudFront CDN distributions, and establish CloudWatch tracing.	• Terraform / IaC blueprints • AWS IAM security keys • Container configuration files

12. ACCELERATED ROADMAP & STRATEGIC EXPANSION

To compress development and guarantee a hardened, production-ready live release by **August 10, 2026**, teams will operate on a highly accelerated weekly sprint schedule.

12.1 Deep Sprint Breakdown Matrix

Sprint Timeline	Core Engineering & Functional Tasks	Target Milestone Gate
June 17 – June 30	Initialize central web repositories. Configure Next.js baseline boilerplates. Provision AWS VPC infrastructure, configure IAM least-privilege policies, and freeze relational Aurora database schemas.	Cloud & Database Infrastructure Freeze
July 01 – July 15	Build Node.js endpoints for experience point updates and leaderboard routing. Package core microservices into Docker images and deploy onto AWS EKS. Configure font renderer strings.	EKS Containers & Gamification Active
July 16 – July 25	Implement time-proctored test engines with 100:15 database shuffling. Inject frontend anti-copy blocks and visibility state listeners. Establish CloudWatch transaction monitoring.	Examination Integrity & Tracing Completed
July 26 – August 10	Integrate generative AI model endpoints via AWS Bedrock and Amazon OpenSearch vector spaces. Run end-to-end web browser smoke testing, verify payment webhook processing, and point live Route 53 domain routing.	Official Launch: August 10, 2026

12.2 Long-Term Strategic Horizons (Post-Launch Expansion Phases)

Following the initial release, Phase I (Q4 2026) introduces audio-native synthesis models to transform recorded voice explanations into adaptive learning material. Phase II (H1 2027) deploys

dedicated B2B portal interfaces for university registrars to adjust leaderboard weights. Phase III (Q4 2027) expands globally distributed performance nodes to maintain sub-50ms query speeds across Tier-3 network conditions.